Yunyang Li

Curriculum Vitae/Oct. 2026

CONTACT

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RESEARCH INTERESTS

physical cosmology; cosmic microwave background; gravitational lensing; large-scale structures; statistics; high-performance computing

PROFESSIONAL APPOINTMENTS

Postdoctoral Fellow — KICP, University of Chicago, 2024 –
Graduate Research Assistant — Johns Hopkins University, 2020–2024
Research Assistant — Peking University, 2017–2018

EDUCATION

Ph.D. Astrophysics. Johns Hopkins University. Thesis advisor: Tobias A. Marriage — 2018–2024
B.Sc. Physics. Peking University. Thesis advisor: Joel N. Bregman — 2013–2017

COLLABORATION

BICEP/Keck Collaboration — 2025 –
South Pole Telescope (SPT-3G) Collaboration — 2024 –
Cosmology Large Angular Scale Surveyor (CLASS) Collaboration — 2018 –

REFERENCES

John E. Carlstrom | Professor | carlstrom@uchicago.edu | 2024 – — Univ. of Chicago
W.L. Kimmy Wu | Assistant Professor | wlw@caltech.edu | 2024 – — Caltech
Tobias A. Marriage | Professor | marriage@jhu.edu | 2018 – — Johns Hopkins Univ.
Charles L. Bennett | Professor | cbennett@jhu.edu | 2018 – — Johns Hopkins Univ.

SELECTED AWARDS

KICP Postdoctoral Fellowship — University of Chicago, 2024
May-the-Fourth Scholarship — Peking University, 2016
CSC Visiting Fellowship — China Scholarship Council, 2016
Academic Excellence Award — Peking University, 2016

TALKS

CMB Lensing Deep and Wide — Seminar, UT Austin, 2026
Largest angular scale CMB E-mode polarization measurement with CLASS — ARC Seminar, UIUC, 2025
Largest angular scale CMB E-mode polarization measurement with CLASS — mm Universe, Chicago, 2025
Quest for the large-scale CMB polarization — IHEP Seminar, IHEP/CAS, 2025
Quest for the large-scale CMB polarization — KIAA Seminar, PKU, 2025
Measurement of the Largest-scale E-mode polarization with CLASS — KIPAC Seminar, SLAC, 2025
Measurement of the Largest-scale E-mode polarization with CLASS — Gravity Group Seminar, Princeton, 2025

Polarimetry with the CLASS telescope — KICP Chalk talk, Chicago, 2024
Efficacy of the Variable-delay Polarization Modulator: Lessons Learned from CLASS — CMB-S4 Summer Collaboration Meeting, UIUC, 2024
Towards recovery of the largest-scale polarization with CLASS — AAS 243 Dissertation Talk, New Orleans, 2024
Accessing the largest-scale microwave polarization with CLASS — Dicke Candidate Seminar, Princeton, 2023
CLASS: Recent Progress and the 40 GHz Results — FPD Seminar, SLAC, 2023
CLASS: Recent Progress and the 40 GHz Results — W&C Seminar, JHU, 2023
CLASS: Recent Progress and the 40 GHz Results — Cosmology Group, CCA, 2023
Atmospheric Cloud Polarization — Cosmo Group, Princeton, 2023
Constraints on CMB Temperature evolution with SZ clusters — W&C Seminar, JHU, 2020
The properties of the Galactic hot gaseous halo from X-ray emission — AAS 231, DC, 2018
Is Calvera a strangeon star? — AAS 231, DC, 2018
The hot gaseous halo of the Milky Way including radiative transfer effects — AAS 229, Grapevine, 2017

MEDIA

Seeing the Big Bang's Cosmic Background — NSF Podcast
Light from dawn of the universe observed by Earth-based telescopes — NSF Stories
Earth-based telescopes offer a fresh look at cosmic dawn — Johns Hopkins Hub

TEACHING

Introduction to Galaxies and Active Galactic Nuclei (TA) — Johns Hopkins Univ., Spring, 2020
General Physics/Lab (TA) — Johns Hopkins Univ., Spring, 2019

OUTREACH

Guest astronomer at the Adler Planetarium (50+ hours) — Chicago, IL, 2024-
Johns Hopkins Physics Fair — Baltimore, MD, 2019

PUBLICATIONS

AS MAIN AUTHOR

7. A Measurement of the Largest-scale CMB E-mode Polarization with CLASS
   ApJ 986, 111 (2025) | **YL** & CLASS Collaboration
6. CLASS Angular Power Spectra and Map-component Analysis for 40 GHz Observations through 2022
   ApJ 963, 92 (2024) | Eimer, Joseph R.; **YL** & CLASS Collaboration
5. CLASS Observations of Atmospheric Cloud Polarization at millimeter Wavelengths
   ApJ 958, 154 (2023) | **YL** & CLASS Collaboration
4. CLASS Data Pipeline and Maps for 40 GHz Observations through 2022
   ApJ 956, 77 (2023) | **YL** & CLASS Collaboration
3. Constraining Cosmic Microwave Background Temperature Evolution With Sunyaev-Zel'Dovich Galaxy Clusters from the Atacama Cosmology Telescope
   ApJ 922, 136 (2021) | **YL**; Hincks, Adam D. & ACT Collaboration
2. Calvera: A Low-mass Strangeon Star Torqued by Debris Disk?
   ApJ 854, 165 (2018) | **YL**; Wang, Weiyang; Ge, Mingyu; Liu, Xiongwei; Tong, Hao & Xu, Renxin
1. The Properties of the Galactic Hot Gaseous Halo from X-Ray Emission
   ApJ 849, 105 (2017) | **YL** & Bregman, Joel

AS CONTRIBUTING AUTHOR

8. Improved Constraints on Cosmic Microwave Background Circular Polarization with CLASS
arXiv 2609.40125 (2026) | Essinger-Hileman, Thomas; Ryan, Caleigh; **YL**; Petroff, Matthew A. & CLASS Collaboration

7. CMB Lensing Reconstruction Using Two Years of Temperature Data from the SPT-3G Summer Survey
arXiv 2607.05784 (2026) | Levy, K.; Raghunathan, S.; Guidi, F.; **YL** & SPT-3G Collaboration

6. Cosmology Large Angular Scale Surveyor (CLASS): 90 GHz Telescope Pointing, Beam Profile, Window Function, and Polarization Performance
ApJS 273, 26 (2024) | Datta, Rahul; Brewer, Michael K.; Denes Couto, Jullianna; Eimer, Joseph; **YL** & CLASS Collaboration

5. Two-year Cosmology Large Angular Scale Surveyor (CLASS) Observations: 40 GHz Telescope Pointing, Beam Profile, Window Function, and Polarization Performance
ApJ 891, 134 (2020) | Xu, Zhilei; Brewer, Michael K.; Fluxá Rojas, Pedro; **YL** & CLASS Collaboration

4. Two-year Cosmology Large Angular Scale Surveyor (CLASS) Observations: A Measurement of Circular Polarization at 40 GHz
ApJ 889, 105 (2020) | Padilla, Ivan L.; Eimer, Joseph R.; **YL** & CLASS Collaboration

3. The Extended Distribution of Baryons around Galaxies
ApJ 862, 3 (2018) | Bregman, Joel N.; Anderson, Michael E.; Miller, Matthew J.; Hodges-Kluck, Edmund; Dai, Xinyu; Li, Jiang-Tao; **YL** & Qu, Zhijie

2. The optical/UV excess of X-ray-dim isolated neutron star II. Nonuniformity of plasma on a strangeon star surface
RAA 18, 082 (2018) | Wang, Wei-Yang; Feng, Yi; Lai, Xiao-Yu; **YL**; Lu, Ji-Guang; Chen, Xuelei & Xu, Ren-Xin

1. Merging strangeon stars
RAA 18, 024 (2018) | Lai, Xiao-Yu; Yu, Yun-Wei; Zhou, En-Ping; **YL** & Xu, Ren-Xin

AS COLLABORATION AUTHOR

25. SPT-3G D1: a measurement of secondary cosmic microwave background anisotropy power
JCAP 2026, 033 (2026) | Chaubal, P.; Huang, N.; Reichardt, C. L. & **SPT-3G Collaboration**

24. Constraints on Millimeter-wave Extragalactic Transient Populations from 4 Years of SPT-3G data
arXiv 2610.28633 (2026) | Guns, S.; Tandoi, C. & **SPT-3G Collaboration**

23. Measuring the Clustering of tSZ-Selected Galaxy Clusters with SPT-3G
arXiv 2610.08947 (2026) | Martsen, E. S.; Anbajagane, D.; Bleem, L. E.; Benson, B. A. & **SPT-3G Collaboration**

22. SPT-3G D1: Foreground-Robust Lensing Templates for Primordial Gravitational Wave Searches
arXiv 2608.06343 (2026) | Nakato, Y.; Wu, W. L. K.; Omori, Y. & **SPT-3G Collaboration**

21. SPT-3G D1: Quadratic-Estimator CMB Lensing Reconstruction and Cosmology
arXiv 2608.31136 (2026) | Omori, Y.; Wu, W. L. K.; Nakato, Y.; Bianchini, F.; Balkenhol, L.; Daley, C.; Quan, W. & **SPT-3G Collaboration**

20. Towards constraining cosmological parameters with SPT-3G observations of 25% of the sky
PhRvD 114, 043548 (2026) | Vitrier, A.; Fichman, K.; Balkenhol, L.; Camphuis, E.; Guidi, F. & **SPT-3G Collaboration**

19. Galaxy Clusters Selected via the Sunyaev-Zel'dovich Effect in 5 year data from the SPT-3G Main Survey
arXiv 2607.01175 (2026) | Bleem, L. E.; Klein, M.; Kornoelje, K.; Bocquet, S.; Sobrin, J. A. & **SPT-3G Collaboration**

18. Characterization of the Polarization Beam Response of SPT-3G Using Point Sources
OJAp 9, 65326 (2026) | de Haan, T.; Archipley, M.; Huang, N. & **SPT-3G Collaboration**

17. SPT-3G D1: Compton-y maps using data from the SPT-3G and Planck surveys
PhRvD 114, 023534 (2026) | Maniyar, A. S.; Bianchini, F.; Wu, W. L. K.; Raghunathan, S. & **SPT-3G Collaboration**

16. Polarization Angle Swings in Blazars Detected in the Millimeter-wave with the South Pole Telescope
arXiv 2608.00302 (2026) | Simpson, A. & **SPT-3G Collaboration**

15. Cross-correlation of SPT-3G D1 CMB lensing and DES Y3 galaxy lensing
arXiv 2606.26223 (2026) | Ouellette, A.; Omori, Y. & **SPT-3G Collaboration**

14. Millimeter-wave Detections of Symbiotic Stars in SPT and ACT Data
arXiv 2605.01022 (2026) | Tandoi, C.; Foster, A.; Maccarone, T. J. & **SPT-3G Collaboration**

13. Measurement of the Full Shape of the Thermal Sunyaev-Zel'dovich Power Spectrum from the South Pole Telescope and Herschel-SPIRE Observations

ApJL 1002, L22 (2026) | Raghunathan, S. & **SPTpol/SPT-3G Collaboration**

12. SPT-3G D1: Maps of the millimeter-wave sky from 2019 and 2020 observations of the SPT-3G Main field
arXiv 2603.20163 (2026) | Quan, W.; Camphuis, E.; Daley, C.; Huang, N.; Omori, Y.; Guidi, F.; Anderes, E. & **SPT-3G Collaboration**

11. SPT-3G D1: A Measurement of Secondary Cosmic Microwave Background Anisotropy Power
arXiv 2601.20551 (2026) | Chaubal, P.; Huang, N.; Reichardt, C. L. & **SPT-3G Collaboration**

10. SPT-3G D1: Constraints on inflationary gravitational waves with two years of SPT-3G data
PhRvD 112, 123520 (2025) | Zebrowski, J. A.; Reichardt, C. L. & **SPT-3G Collaboration**

9. PROTOCALC, a W-band Polarized Calibrator for Cosmic Microwave Background Telescopes: Application to Simons Observatory and CLASS
ApJS 279, 30 (2025) | Coppi, Gabriele; Dachlythra, Nadia; Nati, Federico; Dünner-Planella, Rolando; Adler, Alexandre E.; Errard, Josquin; Galitzki, Nicholas; **YL**; Petroff, Matthew A.; Simon, Sara M.; Sang, Ema Tsang King; Aguilar, Amalia Villarrubia; Wollack, Edward J. & Zannoni, Mario

8. High-efficiency and Low-noise Detectors for the Upgraded CLASS 90 GHz Focal Plane
ApJS 278, 56 (2025) | Núñez, Carolina & **CLASS Collaboration**

7. HoverCal + PoloCalC: precise on-flight metrology performance results over CLASS telescope
SPIE 1310217 (2024) | Dünner, Rolando; Astori, Federico; Coppi, Gabriele; Eimer, Joseph; Errard, Josquin; Fluxá, Juan; **YL**; Mezzanzanica, Noemi; Nati, Federico; Petroff, Matthew; Puddu, Roberto; Rojas, Matías & Zannoni, Mario

6. Sensitivity-improved Polarization Maps at 40 GHz with CLASS and WMAP Data
ApJ 971, 41 (2024) | Shi, Rui & **CLASS Collaboration**

5. Microwave Observations of Venus with CLASS
PSJ 4, 154 (2023) | Dahal, Sumit & **CLASS Collaboration**

4. Testing Cosmic Microwave Background Anomalies in E-mode Polarization with Current and Future Data
ApJ 945, 79 (2023) | Shi, Rui & **CLASS Collaboration**

3. Calibration of Transition-edge Sensor (TES) Bolometer Arrays with Application to CLASS
ApJS 262, 52 (2022) | Appel, John W. & **CLASS Collaboration**

2. Four-year Cosmology Large Angular Scale Surveyor (CLASS) Observations: On-sky Receiver Performance at 40, 90, 150, and 220 GHz Frequency Bands
ApJ 926, 33 (2022) | Dahal, Sumit & **CLASS Collaboration**

1. Two Year Cosmology Large Angular Scale Surveyor (CLASS) Observations: Long Timescale Stability Achieved with a Front-end Variable-delay Polarization Modulator at 40 GHz
ApJ 922, 212 (2021) | Harrington, Kathleen & **CLASS Collaboration**